

MD ARAFAT HOSSAIN

Staff Engineer | iOS, macOS

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CAREER SUMMARY

Experienced Software Engineer with 13+ years building scalable iOS, macOS, Windows, and IoT applications. Contributed to large-scale products including SmartThings iOS (62M+ users) and led core development for Samsung DeX for PC on macOS. Skilled in Swift, Swift Concurrency, UIKit/SwiftUI, MVVM/MVVM-C, IPC, and cross-platform architecture. Recently focused on embedded Godot Engine integration for SmartThings 3D experiences and contributing to SwiftGodotKit/Godot iOS ecosystem development.

TECHNICAL SKILLS

- Swift & Swift 6 (Concurrency – async/await, actors), UIKit, SwiftUI, Objective-C
- Combine, ReactiveSwift, MVVM, MVVM-C, Redux
- RESTful API integration, XCTest, Time Profiler
- Embedded rendering integration (Godot Engine)
- Cross-platform architecture & IPC
- macOS notarization & release pipeline
- Open-source contribution workflows
- Git, GitHub, GitHub Actions, Jenkins, Perforce
- C/C++, C#, Java, WPF, MFC, Android (Java)

WORK EXPERIENCE

Samsung R&D Institute Bangladesh Ltd. | Staff Engineer-I | Mar 2021 – Present

SmartThings iOS

- Led development of Settings and MapView features for SmartThings iOS (62M+ active users, 4.5+ App Store rating).
- Integrated new device cards (TV, lights, sensors) and enhanced MapView user experience.
- Migrated modules from ReactiveSwift to Combine and modernized application architecture.
- Migrating existing modules to Swift 6 Concurrency (async/await, actors).
- Conducted performance optimization using Time Profiler and resolved major market VoCs.

Godot Migration for MapView 3D

- Exploring embedded Godot Engine integration within SmartThings iOS to replace Unity-based rendering workflows.
- Contributed to SwiftGodotKit and Godot iOS embedding support for native Swift ↔ Godot interoperability.
- Worked on native-engine communication and rendering lifecycle integration for iOS.

Other Key Contributions

- Modernized UI and improved IPC for Samsung Medison 5D Viewer (WPF) used in medical ultrasound systems.
- Technical Lead for Samsung DeX for PC (7M+ downloads, 757K+ monthly users), contributing to planning, architecture, and optimization.
- Performed memory optimization and code reviews for MFC-based Windows applications.
- Achieved Professional Level (Top 20%) in Samsung's SW Certificate Test.
- Awarded Star Employee of the Month (January 2026).

Samsung R&D Institute Bangladesh Ltd. | Engineer-III | Mar 2019 – Feb 2021

- Lead macOS engineer for Samsung DeX for PC, owning architecture, release pipeline, notarization, and Apple Silicon transition.
- Delivered universal macOS application support for Apple Silicon (M1).

- Implemented USB and wireless connectivity between Android devices and macOS.
- Developed file transfer, RTSP-based screen rendering, and input-sharing features.
- Ensured performance, stability, and cross-platform compatibility.

Samsung R&D Institute Bangladesh Ltd. | Engineer-II | Mar 2017 – Feb 2019

- Built UI Automator solutions for Android and Tizen platforms for internal B2B device testing.
- Developed an iOS UI automation framework similar to Appium.
- Created an Eclipse plugin to generate automated test cases and integrate internal tools.

Samsung R&D Institute Bangladesh Ltd. | Engineer-I | Jun 2013 – Feb 2017

- Contributed to [IoTivity SDK](#), adopted by Samsung, LG, and Microsoft IoT solutions.
- Developed [OCF](#)-compliant APIs enabling cross-platform IoT interoperability.
- Built a C# .NET desktop application to remotely control Android & Tizen devices.
- Achieved **Advanced** Level (Top 50%) in Samsung's **SW Certificate Test**.

RTC Hubs Limited | Junior Programmer | Sep 2012 – Jun 2013

- Created Multi Shot Cam, a top-charted iOS app generating highest revenue at RTC Hubs.
- Developed 8+ iOS apps including camera & productivity tools using [GPUImage](#).

MENTORING AND LEADERSHIP

SmartThings (iOS) – Settings & MapView | Mar 2024 – Present

- **Part Lead** for Settings & MapView; owned roadmap, design, and delivery for a **6-member** team.
- Mentoring engineers on Swift, SwiftUI/UIKit, MVVM, Combine, and development best practices.
- Supporting feature planning, technical discussions, and cross-team collaboration.
- Coordinating issue resolution with QA and Product teams for high-priority market VoCs.

Samsung Medison – 5D Viewer | Jan 2023 – Feb 2024

- **Tech Lead** overseeing documentation, cross-team collaboration, and knowledge sharing.
- Facilitated onboarding and architecture clarification for distributed teams.

Samsung DeX for PC | Mar 2019 – Dec 2022

- **Tech Lead** for Windows/macOS/Android teams; ensured high-quality feature delivery.
- Mentored **12+ engineers** and guided architecture, performance, and release pipeline.
- Delivered PoC to Samsung CTO securing executive approval.
- Collaborated with PMs and VPs to define functional/non-functional requirements and drive product direction.

Organization-wide Contributions

- Onboarded 7+ engineers across multiple projects.
- Participated in technical interviews and candidate evaluations.
- Conducted internal technical training sessions and knowledge-sharing activities.

Open Source Contributions

- Contributor to [SwiftGodotKit](#) and Godot iOS embedding ecosystem for native Swift interoperability and [embedded rendering](#) workflows.
- Contributed fixes and improvements related to Swift ↔ Godot integration for iOS platforms.

EDUCATION AND CERTIFICATION

B.Sc. in Computer Science and Engineering

Rajshahi University of Engineering & Technology

Feb 2008 – Sep 2012